

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12416242
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Lutjen; Paul M. et al.

Blade outer air seal with compliant seal

Abstract

A gas turbine engine includes a turbine section that has a ceramic matrix composite body that defines a core gaspath side that a row of blades, a forward end, an aft end, and circumferential sides. The forward end abuts at least one vane of a first row of vanes, and the aft end including a seal channel. A compliant seal is disposed in the seal channel and seals against at least one vane of a second row of vanes.

Inventors: Lutjen; Paul M. (Kennebunkport, ME), Rogers; Daniel S. (Lyman, ME), Harding; Timothy J. (Wethersfield, CT), Weisheit; James E. (Kennebunk, ME), Sherman; Morely S. (Biddeford, ME), Rogers; Mikayla M. (Portland, ME), Thayer; Marshall F. (Waterboro, ME), Micucci; Joseph (Scarborough, ME)

Applicant: RTX CORPORATION (Farmington, CT)

Family ID: 1000008820106

Assignee: RTX CORPORATION (Farmington, CT)

Appl. No.: 18/456787

Filed: August 28, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240110488 A1	Apr. 04, 2024

Related U.S. Application Data

us-provisional-application US 63411893 20220930

Publication Classification

Int. Cl.: F01D11/16 (20060101); **F01D11/00** (20060101); **F01D11/08** (20060101)

U.S. Cl.:

CPC **F01D11/16** (20130101); **F01D11/005** (20130101); **F01D11/08** (20130101); F05D2240/55 (20130101); F05D2260/38 (20130101); F05D2300/6033 (20130101)

Field of Classification Search

CPC: F01D (11/16); F01D (11/08); F01D (11/005); F05D (2240/55); F05D (2260/38); F05D (2300/6033)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6821085	12/2003	Darkins, Jr.	415/173.1	F01D 11/08
8079807	12/2010	Shapiro et al.	N/A	N/A
8661828	12/2013	Pieussergues	60/753	F02C 7/28
9874104	12/2017	Shapiro	N/A	F01D 11/08
10392967	12/2018	Correia et al.	N/A	N/A
10422239	12/2018	Mitchell	N/A	F16J 15/32
10487678	12/2018	Tyler, Jr. et al.	N/A	N/A
11028712	12/2020	Tyler et al.	N/A	N/A
11085314	12/2020	Davis	N/A	F01D 25/246
11773751	12/2022	Freeman	415/173.1	F01D 11/08
2010/0111678	12/2009	Habarou et al.	N/A	N/A
2013/0113168	12/2012	Lutjen et al.	N/A	N/A
2017/0130600	12/2016	Shapiro et al.	N/A	N/A
2021/0189909	12/2020	Sippel	N/A	F01D 11/08

OTHER PUBLICATIONS

European Search Report for European Patent Application No. 23201303.7 mailed Feb. 15, 2024. cited by applicant

Primary Examiner: Wiehe; Nathaniel E

Assistant Examiner: Ribadeneyra; Theodore C

Attorney, Agent or Firm: Carlson, Gaskey & Olds, P.C.

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) The present disclosure claims priority to U.S. Provisional Application No. 63/411,893 filed Sep. 30, 2022.

BACKGROUND

(1) A gas turbine engine typically includes a fan section, a compressor section, a combustor section and a turbine section. Air entering the compressor section is compressed and delivered into the

combustion section where it is mixed with fuel and ignited to generate a high-pressure and temperature exhaust gas flow. The high-pressure and temperature exhaust gas flow expands through the turbine section to drive the compressor and the fan section. The compressor section may include low and high pressure compressors, and the turbine section may also include low and high pressure turbines.

(2) Components in the turbine section are typically formed of a superalloy and may include thermal barrier coatings to extend temperature capability and lifetime. Ceramic matrix composite (“CMC”) materials are also being considered for turbine components. Among other attractive properties, CMCs have high temperature resistance. Despite this attribute, however, there are unique challenges to implementing CMCs in turbine components.

SUMMARY

(3) A gas turbine engine according to an example of the present disclosure includes a turbine section that has a first row of vanes disposed about an engine axis, a second row of vanes axially spaced from the first row of vanes, and a row of blades axially between the first row of vanes and the second row of vanes. There is a row of blade outer air seals axially between the first row of vanes and the second row of vanes and that is situated radially outwardly of the row of blades. Each blade outer air seal has a ceramic matrix composite body that defines a core gaspath side facing the row of blades, a forward end, an aft end, and circumferential sides. The forward end abuts at least one of the vanes of the first row of vanes. The aft end has a seal channel. A compliant seal is disposed in the seal channel and seals against at least one of the vanes of the second row of vanes.

(4) In a further embodiment of any of the foregoing embodiments, the compliant seal is a spring seal.

(5) In a further embodiment of any of the foregoing embodiments, the spring seal is a W-seal.

(6) In a further embodiment of any of the foregoing embodiments, the compliant seal is a rope seal.

(7) In a further embodiment of any of the foregoing embodiments, the ceramic matrix composite body defines a non-core gaspath side and includes a pair of flanges that projects from the non-core gaspath side, and the pair of flanges includes first and second pin holes that are aligned to receive a pin.

(8) In a further embodiment of any of the foregoing embodiments, the aft end of the ceramic matrix composite body includes radially inner and outer walls that define the seal channel therebetween.

(9) In a further embodiment of any of the foregoing embodiments, the compliant seal is a full hoop around the engine axis.

(10) In a further embodiment of any of the foregoing embodiments, thermal expansion and thermal contraction of the least one of the vanes of the first row of vanes that is abutting the blade outer air seal causes axial movement of the blade outer air seal to compress and decompress, respectively, the compliant seal.

(11) In a further embodiment of any of the foregoing embodiments, the vanes of the first row of vanes abut each of the blade outer air seals at a respective face seal, and the compliant seal biases the blade outer air seal in an axial forward direction to maintain contact in the face seal.

(12) A blade outer air seal according to an example of the present disclosure includes a ceramic matrix composite body that defines a core gaspath side, a forward end, an aft end, and circumferential sides. The aft end has a seal channel. A compliant seal is disposed in the seal channel.

(13) In a further embodiment of any of the foregoing embodiments, the compliant seal is a spring seal.

(14) In a further embodiment of any of the foregoing embodiments, the spring seal is a W-seal.

(15) In a further embodiment of any of the foregoing embodiments, the compliant seal is a rope seal.

(16) In a further embodiment of any of the foregoing embodiments, the ceramic matrix composite

body defines a non-core gaspath side and includes a pair of flanges that projects from the non-core gaspath side, and the pair of flanges includes first and second pin holes that are aligned to receive a pin.

(17) In a further embodiment of any of the foregoing embodiments, the aft end of the ceramic matrix composite body includes radially inner and outer walls that define the seal channel therebetween.

(18) The present disclosure may include any one or more of the individual features disclosed above and/or below alone or in any combination thereof.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The various features and advantages of the present disclosure will become apparent to those skilled in the art from the following detailed description. The drawings that accompany the detailed description can be briefly described as follows.

(2) FIG. 1 illustrates a gas turbine engine.

(3) FIG. 2 illustrates a turbine section from the engine of FIG. 1.

(4) FIG. 3 illustrates the turbine section with a different kind of compliant seal.

(5) FIG. 4 illustrates a full hoop seal, a seal with a circumferential split, or a multiply seal with an overlapping shiplap split section.

(6) In this disclosure, like reference numerals designate like elements where appropriate and reference numerals with the addition of one-hundred or multiples thereof designate modified elements that are understood to incorporate the same features and benefits of the corresponding elements.

DETAILED DESCRIPTION

(7) FIG. 1 schematically illustrates a gas turbine engine 20. The gas turbine engine 20 is disclosed herein as a two-spool turbofan that generally incorporates a fan section 22, a compressor section 24, a combustor section 26 and a turbine section 28. The fan section 22 drives air along a bypass flow path B in a bypass duct defined within a housing 15 such as a fan case or nacelle, and also drives air along a core flow path C for compression and communication into the combustor section 26 then expansion through the turbine section 28. Although depicted as a two-spool turbofan gas turbine engine in the disclosed non-limiting embodiment, it should be understood that the concepts described herein are not limited to use with two-spool turbofans as the teachings may be applied to other types of turbine engines including three-spool architectures.

(8) The exemplary engine 20 generally includes a low speed spool 30 and a high speed spool 32 mounted for rotation about an engine central longitudinal axis A relative to an engine static structure 36 via several bearing systems 38. It should be understood that various bearing systems 38 at various locations may alternatively or additionally be provided, and the location of bearing systems 38 may be varied as appropriate to the application.

(9) The low speed spool 30 generally includes an inner shaft 40 that interconnects, a first (or low) pressure compressor 44 and a first (or low) pressure turbine 46. The inner shaft 40 is connected to the fan 42 through a speed change mechanism, which in exemplary gas turbine engine 20 is illustrated as a geared architecture 48 to drive a fan 42 at a lower speed than the low speed spool 30. The high speed spool 32 includes an outer shaft 50 that interconnects a second (or high) pressure compressor 52 and a second (or high) pressure turbine 54. A combustor 56 is arranged in the exemplary gas turbine 20 between the high pressure compressor 52 and the high pressure turbine 54. A mid-turbine frame 57 of the engine static structure 36 may be arranged generally between the high pressure turbine 54 and the low pressure turbine 46. The mid-turbine frame 57 further supports bearing systems 38 in the turbine section 28. The inner shaft 40 and the outer shaft

50 are concentric and rotate via bearing systems **38** about the engine central longitudinal axis **A** which is collinear with their longitudinal axes.

(10) The core airflow is compressed by the low pressure compressor **44** then the high pressure compressor **52**, mixed and burned with fuel in the combustor **56**, then expanded through the high pressure turbine **54** and low pressure turbine **46**. The mid-turbine frame **57** includes airfoils **59** which are in the core airflow path **C**. The turbines **46**, **54** rotationally drive the respective low speed spool **30** and high speed spool **32** in response to the expansion. It will be appreciated that each of the positions of the fan section **22**, compressor section **24**, combustor section **26**, turbine section **28**, and fan drive gear system **48** may be varied. For example, gear system **48** may be located aft of the low pressure compressor, or aft of the combustor section **26** or even aft of turbine section **28**, and fan **42** may be positioned forward or aft of the location of gear system **48**.

(11) The engine **20** in one example is a high-bypass geared aircraft engine. In a further example, the engine **20** bypass ratio is greater than about six (6), with an example embodiment being greater than about ten (10), and can be less than or equal to about 18.0, or more narrowly can be less than or equal to 16.0. The geared architecture **48** is an epicyclic gear train, such as a planetary gear system or other gear system, with a gear reduction ratio of greater than about 2.3. The gear reduction ratio may be less than or equal to 4.0. The low pressure turbine **46** has a pressure ratio that is greater than about five. The low pressure turbine pressure ratio can be less than or equal to 13.0, or more narrowly less than or equal to 12.0. In one disclosed embodiment, the engine **20** bypass ratio is greater than about ten (10:1), the fan diameter is significantly larger than that of the low pressure compressor **44**, and the low pressure turbine **46** has a pressure ratio that is greater than about five 5:1. Low pressure turbine **46** pressure ratio is pressure measured prior to an inlet of low pressure turbine **46** as related to the pressure at the outlet of the low pressure turbine **46** prior to an exhaust nozzle. The geared architecture **48** may be an epicycle gear train, such as a planetary gear system or other gear system, with a gear reduction ratio of greater than about 2.3:1 and less than about 5:1. It should be understood, however, that the above parameters are only exemplary of one embodiment of a geared architecture engine and that the present invention is applicable to other gas turbine engines including direct drive turbofans.

(12) A significant amount of thrust is provided by the bypass flow **B** due to the high bypass ratio. The fan section **22** of the engine **20** is designed for a particular flight condition—typically cruise at about 0.8 Mach and about 35,000 feet (10,668 meters). The flight condition of 0.8 Mach and 35,000 ft (10,668 meters), with the engine at its best fuel consumption—also known as “bucket cruise Thrust Specific Fuel Consumption (‘TSFC’)”—is the industry standard parameter of lbf of fuel being burned divided by lbf of thrust the engine produces at that minimum point. The engine parameters described above and those in this paragraph are measured at this condition unless otherwise specified. “Low fan pressure ratio” is the pressure ratio across the fan blade alone, without a Fan Exit Guide Vane (“FEGV”) system. The low fan pressure ratio as disclosed herein according to one non-limiting embodiment is less than about 1.45, or more narrowly greater than or equal to 1.25. “Low corrected fan tip speed” is the actual fan tip speed in ft/sec divided by an industry standard temperature correction of $[(T_{\text{fan}} \text{ } ^\circ \text{R}) / (518.7 \text{ } ^\circ \text{R})]^{0.5}$. The “Low corrected fan tip speed” as disclosed herein according to one non-limiting embodiment is less than about 1150.0 ft/second (350.5 meters/second), and can be greater than or equal to 1000.0 ft/second (304.8 meters/second).

(13) FIG. 2 illustrates a portion of the high pressure turbine **54** but is also applicable to the low pressure turbine **46**. This section of the turbine **54** includes a first row of vanes **60** and a second row of vanes **62** axially spaced from the first row of vanes **60**. There is a row of rotatable blades **64** axially between the first row of vanes **60** and the second row of vanes **62**. A row of blade outer air seals **66** (“BOAS **66**”) is located axially between the first row of vanes **60** and the second row of vanes **62**.

(14) Each BOAS **66** is situated radially outwardly of the row of blades **64** and is comprised of a

ceramic matrix composite body **68** (“CMC body **68**”). A CMC material is comprised of one or more ceramic fiber plies in a ceramic matrix. Example ceramic matrices are silicon-containing ceramic, such as but not limited to, a silicon carbide (SiC) matrix or a silicon nitride (Si₃N₄) matrix. Example ceramic reinforcement of the CMC are silicon-containing ceramic fibers, such as but not limited to, silicon carbide (SiC) fiber or silicon nitride (Si₃N₄) fibers. The CMC may be, but is not limited to, a SiC/SiC ceramic matrix composite in which SiC fiber plies are disposed within a SiC matrix. A fiber ply has a fiber architecture, which refers to an ordered arrangement of the fiber tows relative to one another, such as a 2D woven ply or a 3D structure.

(15) The CMC body **68** defines a forward end **68a**, an aft end **68b**, circumferential sides **68c**, a core gaspath side **68d** facing the row of blades **64**, and a non-core gaspath side **68e** opposite the core gaspath side **68d**. There is a pair of flanges **70** that project radially from the non-core gaspath side **68e** for mounting the BOAS **66** in the engine **20**. For example, the flanges **70** define pin holes **72** that are aligned to receive a support pin there through. In that regard, a mating support structure **74**, such as an engine case or other hardware, includes complementary flanges **76** and pin holes **78**. A support pin is received through the pin holes **72/78** to secure the BOAS **66** to the support structure **74**. The pinned joint substantially radially supports the BOAS **66** but the joint is non-rigid and permits some movement of the BOAS **66** so as to avoid over-constraint.

(16) The forward end **68a** of the CMC body **68** abuts at least one of the vanes **60** of the first row of vanes **60**. For instance, each of the vanes **60** includes a platform **60a** with a radially-projecting flange **60b** that the CMC body **68** bears against. The joint formed by the contact between the flange **60b** and the CMC body **68** is rigid in that the vane **60** and the BOAS **66** essentially act as a unitary body with respect to axial movement. The contact between flange **60b** and **68a** act as a face seal since they remain in contact. This seal can be a primary seal or used in parallel with another seal.

(17) CMC material in general has lower thermal conductivity than superalloys and does not possess the same strength and ductility characteristics as superalloys, making CMCs more susceptible to distress from aerodynamic and/or other loads and thermally induced stresses caused by thermal gradients. Due to tolerances and thermal growth in gas turbine engines, it is difficult to maintain axial spacing at interfaces between a BOAS and the adjacent forward and aft vanes, thus making it difficult to maintain sealing at those interfaces. Moreover, mechanically constraining components to reduce variations at the interfaces is undesirable for CMCs. In these regard, as explained below, the BOAS **66** includes a compliant seal **80**.

(18) The aft end **68b** of the CMC body **68** includes a seal channel **82** that is defined between radially inner and outer walls **84a/84b** of the CMC body **68**. The compliant seal **80** is disposed in the seal channel **82** and seals against at least one of the vanes **62** of the second row of vanes **62**. For instance, each of the vanes **62** includes a platform **62a** with a radially-projecting flange **62b** that the compliant seal bears against. The joint formed by the contact between the flange **62b** and the compliant seal **80** is non-rigid in that the unitary structure formed by the vane **60** and the CMC body **68** are able to compress and decompress the compliant seal **80**, although the compliant seal **80** will still be compressed to some degree in the “decompressed” state in order to provide springback and maintain pressure against the vanes **60/62**. For instance, as the vane **60** and CMC body **68** thermally expand, the compliant seal **80** is compressed, and as the vane **60** and CMC body **68** thermally contract, the compliant seal **80** somewhat decompresses. During compression and decompression the compliant seal maintains contact, and thus sealing, with the vane **62**, thereby permitting the vane **60** and BOAS **66** to move axially without over-constraint that might otherwise cause distress in the CMC material. Moreover, the springback of the compliant seal **80** facilitates keeping the BOAS **66** in contact with, and thus sealed against, the vane **60**.

(19) In the example shown, the compliant seal **80** is a spring seal, such as but not limited to, a W-seal. In general, a W-seal has a cross-section that is shaped liked the letter “W.” Alternatively, as shown in FIG. 3, the compliant seal **180** is a rope seal. For example, the rope seal **180** is formed of fibers, such as ceramic fibers or metallic fibers. The fibers may be braided, knitted, or woven.

Example ceramic fibers include, but are not limited to, oxide fibers. For instance, the ceramic fibers are NEXTEL fibers, which are composed of Al_2O_3 , SiO_2 , and B_2O_3 . Example metallic fibers include, but are not limited to, nickel alloy or a cobalt alloy fibers. The spring seal may have better springback in comparison to the rope seal, but the rope seal may provide better sealing than the spring seal. Additionally, as the compliant seal **80/180** runs circumferentially along the aft end **68b**, the compliant seal **80/180** may circumferentially span two or more blade outer air seals **66**. For example, as shown in FIG. 4, the compliant seal **80** is a full hoop about the engine axis A and spans all of the blade outer air seals **66**.

(20) Although a combination of features is shown in the illustrated examples, not all of them need to be combined to realize the benefits of various embodiments of this disclosure. In other words, a system designed according to an embodiment of this disclosure will not necessarily include all of the features shown in any one of the Figures or all of the portions schematically shown in the Figures. Moreover, selected features of one example embodiment may be combined with selected features of other example embodiments.

(21) The preceding description is exemplary rather than limiting in nature. Variations and modifications to the disclosed examples may become apparent to those skilled in the art that do not necessarily depart from this disclosure. The scope of legal protection given to this disclosure can only be determined by studying the following claims.

Claims

1. A gas turbine engine comprising: a turbine section including, a core gaspath; a first row of vanes disposed in the core gaspath about an engine axis, a second row of vanes in the core gaspath and axially spaced from the first row of vanes, and a row of blades in the core gaspath axially between the first row of vanes and the second row of vanes, each vane of the first row of vanes including a platform with a radially-projecting flange; and a row of blade outer air seals axially between the first row of vanes and the second row of vanes and situated radially outwardly of the row of blades, each blade outer air seal including, a ceramic matrix composite body defining a core gaspath side facing the row of blades and bounding the core gaspath, a forward end, an aft end, and circumferential sides, the forward end abutting the radially-projecting flange of one of the vanes of the first row of vanes, the aft end including a radially inner wall and a radially outer wall defining radially therebetween a seal channel, the radially inner wall defining, opposite the seal channel, an aft portion of the core gaspath side, and the radially inner wall and the radially outer wall are of equal axial length, and a compliant seal disposed in the seal channel and sealing against at least one of the vanes of the second row of vanes.
2. The gas turbine engine as recited in claim 1, wherein the compliant seal is a spring seal.
3. The gas turbine engine as recited in claim 1, wherein the compliant seal is a W-seal.
4. The gas turbine engine as recited in claim 1, wherein the compliant seal is a rope seal.
5. The gas turbine engine as recited in claim 1, wherein the ceramic matrix composite body defines a non-core gaspath side and includes a pair of flanges that projects from the non-core gaspath side, and the pair of flanges includes first and second pin holes that are aligned to receive a pin.
6. The gas turbine engine as recited in claim 1, wherein the compliant seal is a full hoop around the engine axis.
7. The gas turbine engine as recited in claim 1, wherein thermal expansion and thermal contraction of the least one of the vanes of the first row of vanes that is abutting the blade outer air seal causes axial movement of the blade outer air seal to compress and decompress, respectively, the compliant seal.
8. The gas turbine engine as recited in claim 1, wherein the ceramic matrix composite body is a SiC/SiC ceramic matrix composite in which SiC fiber plies are disposed within a SiC matrix.

